

IC Lab Final 2021 Fall

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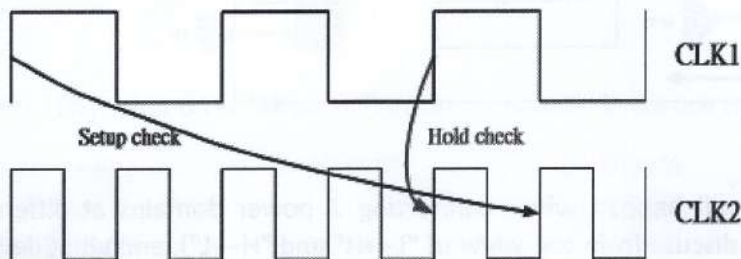
Account: iclab 009

Total score: 100%

1. [4%] Lab07:

Please write two commands relative to "start point" to let the setup time check and hold time check like the picture below :

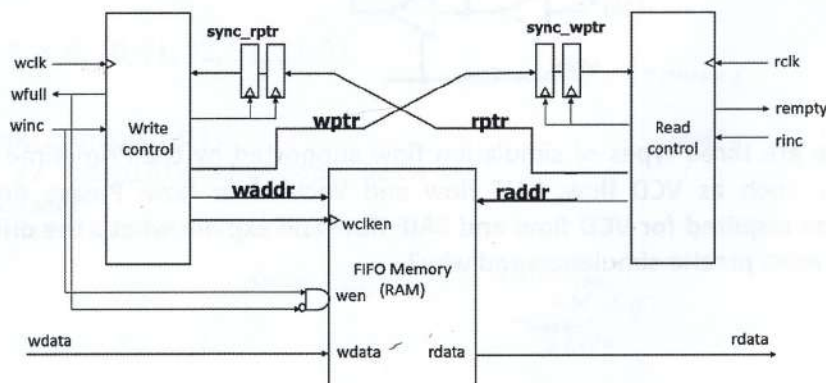
Hint : set_multicycle_path path_multiplier -from \$start_point -to \$end_point [-setup] [-hold] [-start] [-end]



3 start
2 hold
2 start

2. [11%] Lab07:

- [5%] In Dual Clock FIFO synchronizers, why do we need two D-FFs for write pointer? Now given a write action, what is the time relation between the data store in RAM and the read control responding to this action? Please specify number of cycles.
- [3%] Why do we use gray encoding for pointers in Dual Clock FIFO synchronizers ?
- [3%] What is the role of XOR in XOR synchronizer? Please consider both input and output.

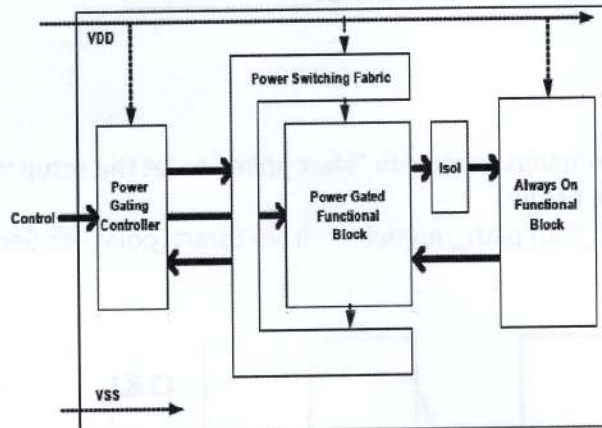


3. [12%] Lab08:

(a) [4%] Below is a classical architecture for power gating design.

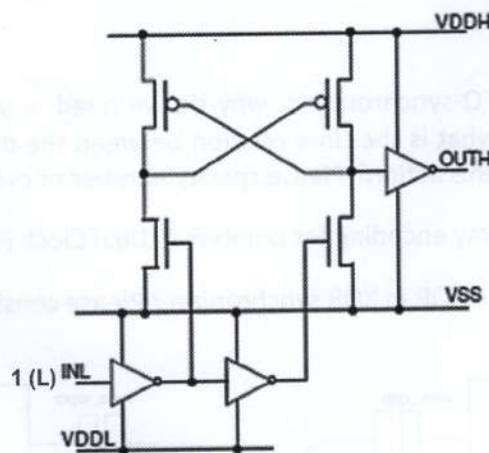
(1) Which logic gate we can use to implement the "isolation cell", how it works and explain why we need it?

(2) Why we need to implement "retention register" in some situations?



(b) [4%] Please describe what will happen when connecting 2 power domains at different voltage levels? (hint: please discuss from the view of "L→H" and "H→L"), and what device we will use to deal with this issue?

level shifter



(c) [4%] There are three types of simulation flow supported by the Primetime tool for power simulation, such as VCD flow, SAIF flow and Vector-free flow. **Please describe files of information required for VCD flow and SAIF flow and explain what's the difference.** Which one is the most precise simulation and why?

VCD *time-based power*
时序图

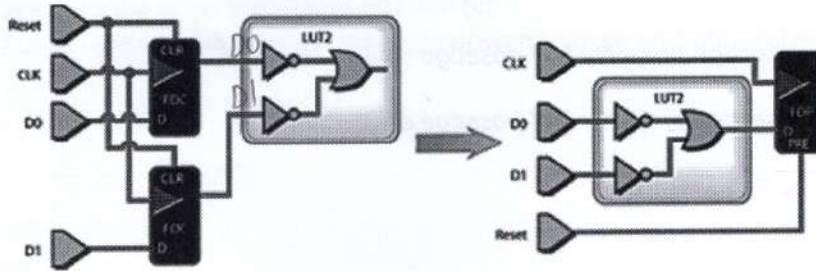
4. [9%] Lab08:

(a) [4%] Please describe the difference between **SEC** (Sequential Equivalence Checking) and **CEC** (Combinational Equivalence Checking)?

(b) [2%] Please answer you will use SEC or CEC in each situation below:

- (1): Compare design after retiming, design after gating.
- (2): Compare RTL design, Gate-level netlist and layout netlist

(c) [2%] If we use the SEC to check the design in figure, will the SEC report equivalent or not?



(d) [1%] What does "**bbox_m**" mean and when will we use this command?

black box

破出内部是一样的/时

why:
当时



5. [6%] Lab09:

In SystemVerilog, why do we generate the pattern by using "**program**" block instead of "**module**"? Please also provide an example to illustrate your answer.

\$display

c = 10; internal

6. [10%] Lab09:

(a) [4%] What is the advantage of using System Verilog? Please list 1 advantage and give a example.

(b) [6%] Please complete the code such that the class random_obj has an **8-bit** member variable named mem whose value is limited within the range below and the random function of mem will exhaust all values before repeating any individual value.

Range = 7, 8, 9, 10, 11, 12, 31, 35, 37

constraint {

```
class random_obj;
    // complete the code here
endclass
```

inside

randc

7. [8%] Lab10:

If we want to check whether our design hits the specification, we can write the assertion in the checker.sv file. There are two signals, A and B in the design, and two signals are triggered when posedge clk. Please finish the two assertions below. Notice that you cannot use always block here.

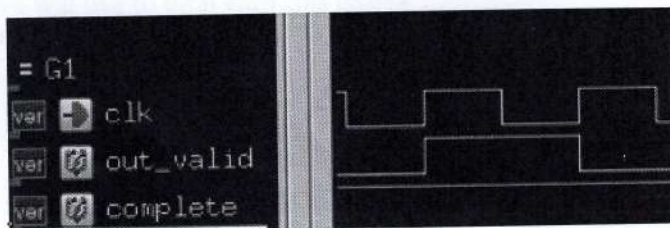
- (a) [4%] If A changes from 0 to 1 between one rising clock, then 2-5 cycles later, B should be high for 5 cycles. (Hint: \$rose() could be used)
- (b) [4%] If A is 0, B should be high 3 clock cycles ago.

Assertion1: assertion property (@(posedge clk)) ;

Assertion2: assertion property (@(posedge clk)) ;

8. [8%] Lab10:

- (a) [6%] If we want to create a covergroup cg1 and sample the value of signal "complete" at negedge clk when the out_valid is high, where the waveforms of out_valid and "complete" are shown below. What is the problem when using the code below? How can we correct it?



```
covergroup cg1@(negedge clk && out_valid);
.....
endgroup
```

- (b) [2%] How can we get the coverage of cg1 during the simulation (without using other EDA tools like imc)?

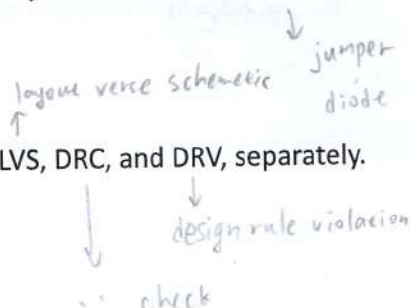
9. [12%] Lab11:

- (a) [6%] The following files: LEF, captbl, CDB, GDS are required for APR, please explain the content or purpose of these files.

- (b) [4%] Please explain what is antenna effect, and list 1 solution for this.

10. [4%] Lab11:

Explain what is LVS, DRC, and DRV, separately.



11. [9%] Lab12:

Please answer the following questions:

- (a) [3%] Please briefly explain what IR drop is. less noise margin
- (b) [3%] Please briefly explain at least 3 reasons why IR drop may cause the chip to fail.
- (c) [3%] Please describe at least 3 methods to prevent IR drop.

12. [7%] Lab12:

- (a) [3%] Please describe what is electromigration (EM).
- (b) [4%] Please describe the difference between power analysis and rail analysis.